IWATANI CORPORATION · CONFIDENTIAL WORKING DRAFT

Metal Bellows
Value Chain Map

Twelve stages from chrome ore to a replacement bellows, with players and technical gates at every stage

Client

Iwatani Corporation — Metals Department → Stainless Steel Division

Subject

Bellows tube (precision-slit material) market entry, leveraging the Jindal Stainless relationship

Status

Working document — updated at monthly internal meetings

Prepared

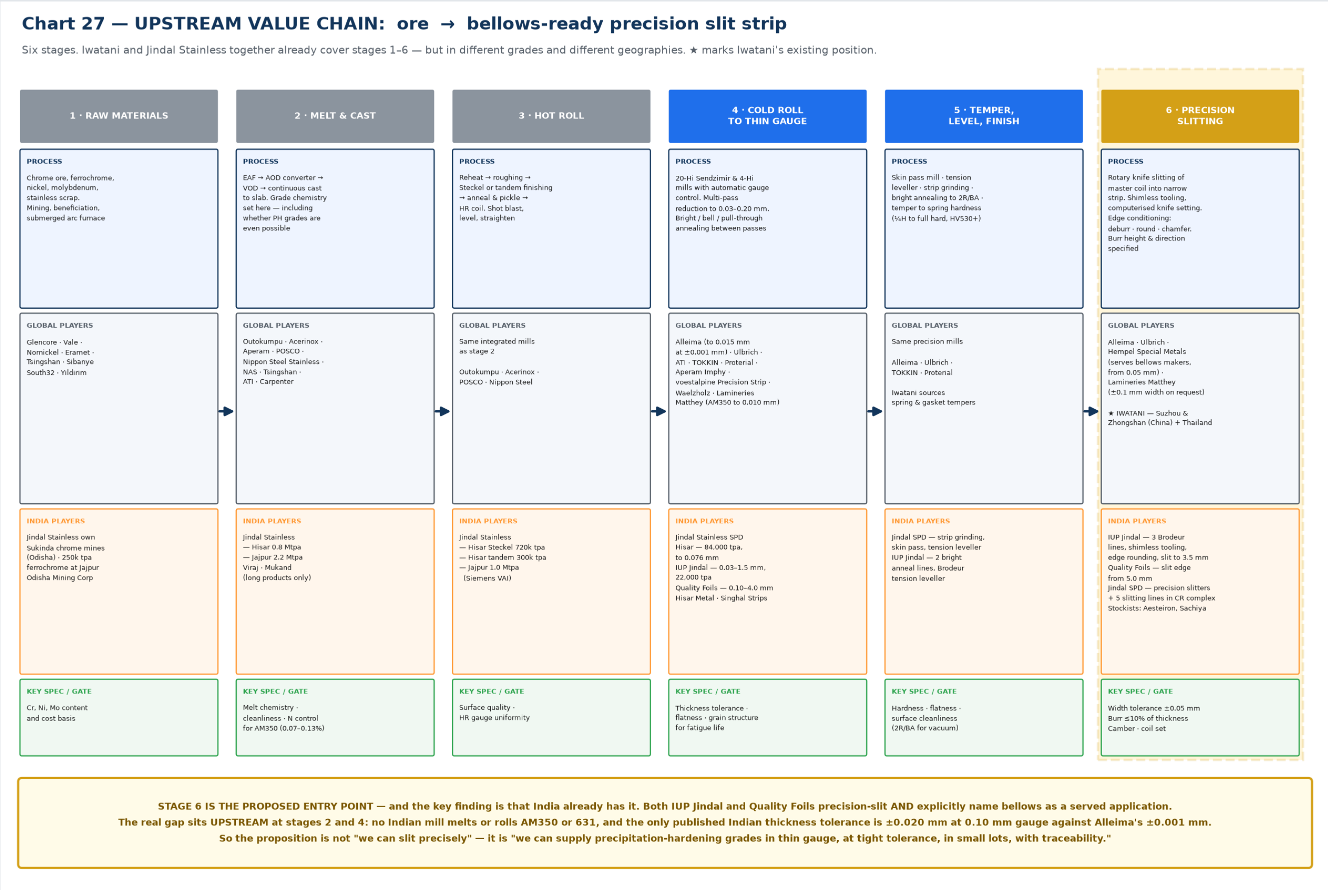
July 2026

Evidence standard. Every factual claim in this document carries an inline source link and a verification grade: **[P]** primary (company, association or government publication), **[C]** credible press, **[S]** commercial market-research vendor (indicative only), **[U]** unverified directory listing. Commercial estimates for the bellows market disagree with one another by up to a factor of 19; figures marked **[S]** must not be quoted externally without stating the range. Landscape charts are best viewed on screen or printed A3.

14 — Metal Bellows Value Chain: Exhaustive Upstream-to-Downstream Map

Twelve stages from chrome ore to a replacement bellows fitted in a running fab. At each stage: what physically happens, who plays globally, who plays in India, and what the technical gate is.

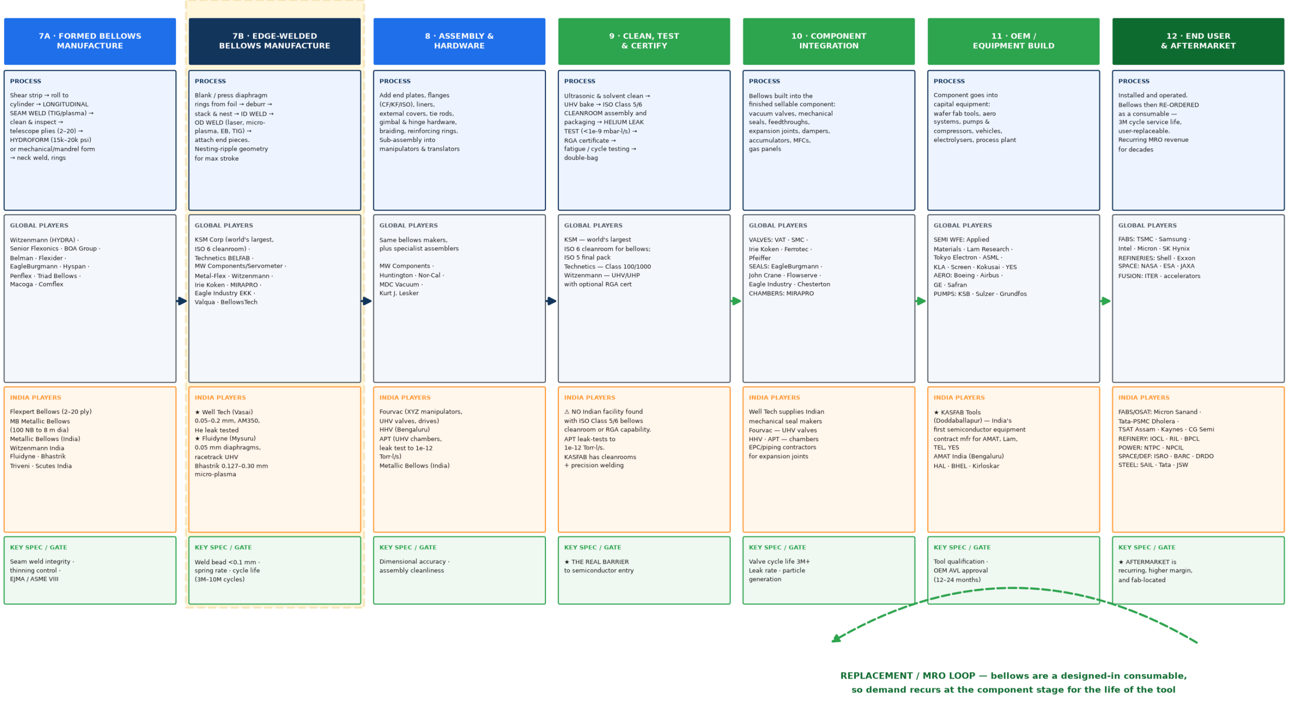
1. The map — upstream



2. The map — midstream and downstream

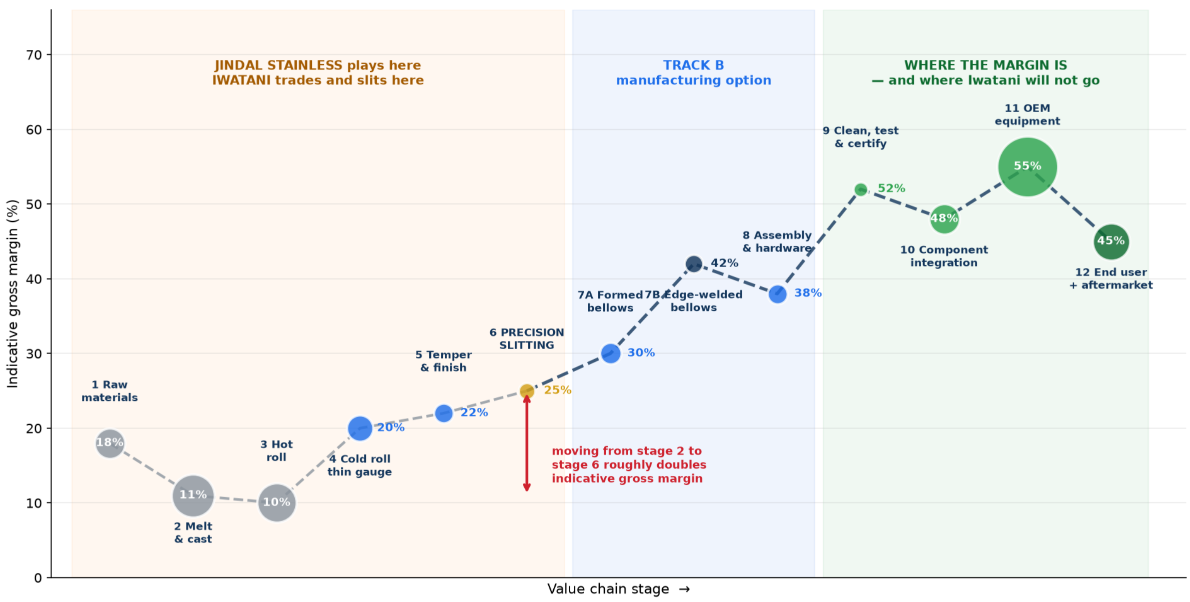
Chart 28 — MIDSTREAM & DOWNSTREAM VALUE CHAIN: precision strip → bellows → component → equipment → end user

Two manufacturing routes diverge at stage 7. ★ marks the Indian players that matter. ⚠ marks the capability India does not have.



3. Where the value sits

Chart 29 — Value pool: margin rises monotonically downstream, and so does the barrier to entry
Bubble size ≈ relative revenue pool. INDICATIVE ESTIMATES — a prioritisation aid, not measured data.



MARGIN FIGURES ARE OUR INDICATIVE ESTIMATES, not measured data, and are shown to make the shape of the chain visible. Bubble areas are relative revenue pools, also indicative. THE SHAPE IS THE POINT: commodity melting and hot rolling sit at 10-11%, precision slitting at ~25%, edge-welded bellows manufacture at ~42%, and cleanroom qualification and component integration at ~48-52%. Every step downstream adds margin AND adds a qualification barrier. Iwatani's realistic ambition is to move from stage 2-3 economics to stage 5-6 economics, which roughly doubles margin without requiring a manufacturing licence, cleanroom or OEM approval. Stages 9-12 are where the money is and are also where a trading house cannot credibly go.

4. Detailed process flow

[Diagram — rendered in the web version. Source below.]

```
flowchart TD
    subgraph UP["UPSTREAM — material"]
        A1["<b>1 RAW MATERIALS</b><br/>Chrome ore · ferrochrome<br/>Ni · Mo · stainless scrap<br/><i>Glencore, Vale, Tsingshan<br/>India: Jindal Sukinda mines</i>"]
        A2["<b>2 MELT & CAST</b><br/>EAF → AOD → VOD →<br/>continuous cast to slab<br/><i>Outokumpu, Acerinox, POSCO<br/>India: Jindal Stainless 3.0 Mtpa</i>"]
        A3["<b>3 HOT ROLL</b><br/>Reheat → rough → finish →<br/>anneal & pickle → HR coil<br/><i>India: Hisar Steckel 720k tpa<br/>Jajpur tandem 1.0 Mtpa</i>"]
        A4["<b>4 COLD ROLL TO THIN GAUGE</b><br/>20-Hi Sendzimir with AGC<br/>to 0.03–0.20 mm<br/><i>Alleima 0.015mm ±0.001mm<br/>TOKKIN · Ulbrich · Matthey<br/>India: Jindal SPD 0.076mm ·<br/>IUP Jindal 0.03mm · Quality Foils</i>"]
        A5["<b>5 TEMPER & FINISH</b><br/>Skin pass · tension level ·<br/>strip grind · bright anneal 2R<br/>temper to HV530+"]
        A6["<b>6 PRECISION SLITTING</b><br/>Rotary knife · shimless tooling<br/>deburr / round / chamfer edge<br/>burr ≤10% of thickness<br/><i>★ Iwatani: Suzhou, Zhongshan<br/>India: IUP Jindal, Quality Foils</i>"]
    end

    A1 --> A2 --> A3 --> A4 --> A5 --> A6

    A6 --> R1
    A6 --> R2

    subgraph MID["MIDSTREAM — bellows manufacture"]
        R1["<b>7A FORMED / CONVOLUTED</b><br/>shear → roll to cylinder →<br/><b>longitudinal seam weld</b> →<br/>clean & inspect → telescope<br/>plies 2–20 → <b>hydroform</b><br/>15–20k psi or mandrel form<br/><i>Witzenmann · Senior · BOA<br/>India: Flexpert · MB · Metallic</i>"]
        R2["<b>7B EDGE-WELDED</b><br/>blank diaphragm rings →<br/>deburr → stack & nest →<br/><b>ID weld</b> → <b>OD weld</b><br/>laser / micro-plasma / EB<br/><i>KSM · Technetics · Irie Koken<br/>India: ★ Well Tech · ★ Fluidyne</i>"]
        R3["<b>8 ASSEMBLY & HARDWARE</b><br/>end plates · CF/KF/ISO flanges<br/>liners · covers · tie rods ·<br/>gimbal/hinge · braiding<br/><i>India: Fourvac · HHV · APT</i>"]
        R4["<b>9 CLEAN, TEST & CERTIFY</b><br/>UHV clean & bake →<br/><b>ISO Class 5/6 cleanroom</b> →<br/><b>helium leak test</b> <1e-9 →<br/>RGA cert → cycle test<br/><i>⚠ NO Indian capability found</i>"]
    end

    R1 --> R3
    R2 --> R3
    R3 --> R4

    subgraph DOWN["DOWNSTREAM — integration to end use"]
        D1["<b>10 COMPONENT INTEGRATION</b><br/>vacuum valves · mechanical seals<br/>feedthroughs · expansion joints<br/>MFCs · gas panels · accumulators<br/><i>VAT · SMC · EagleBurgmann ·<br/>John Crane · MIRAPRO</i>"]
        D2["<b>11 OEM / EQUIPMENT BUILD</b><br/>wafer fab tools · aero systems<br/>pumps · compressors · vehicles<br/>electrolysers · process plant<br/><i>AMAT · Lam · TEL · ASML<br/>India: ★ KASFAB Tools</i>"]
        D3["<b>12 END USER</b><br/>fabs · refineries · power ·<br/>steel · space · defence · H2<br/><i>India: Micron Sanand · Tata<br/>Dholera · TSAT Assam · IOCL ·<br/>NTPC · NPCIL · ISRO</i>"]
    end

    R4 --> D1 --> D2 --> D3
    D3 -.->|"<b>MRO / REPLACEMENT LOOP</b><br/>3M cycle service life ·<br/>user-replaceable · recurring<br/>for the life of the tool"| D1

    style A6 fill:#ffe08a,stroke:#b8860b,stroke-width:3px
    style R2 fill:#cfe8ff,stroke:#1f6feb,stroke-width:3px
    style R4 fill:#ffd7d7,stroke:#cf222e,stroke-width:2px
    style D1 fill:#d7f5d7,stroke:#2da44e
    style D2 fill:#d7f5d7,stroke:#2da44e
    style D3 fill:#d7f5d7,stroke:#2da44e
```

5. Stage-by-stage reference table

#	Stage	Process	Global players	India players	Technical gate
1	Raw materials	Chrome ore, ferrochrome, Ni, Mo, scrap; mining, beneficiation, submerged arc furnace	Glencore, Vale, Nornickel, Eramet, Tsingshan, South32, Yildirim	**Jindal Stainless owns Sukinda chrome mines (Odisha)**, 250k tpa ferrochrome at Jajpur; Odisha Mining Corp	Cr/Ni/Mo content and cost basis
2	Melt & cast	EAF → AOD converter → VOD → continuous cast to slab. **Grade chemistry is fixed here — including whether PH grades are possible at all**	Outokumpu, Acerinox, Aperam, POSCO, Nippon Steel Stainless, NAS, Tsingshan, ATI, Carpenter	**Jindal Stainless** — Hisar 0.8 Mtpa + Jajpur 2.2 Mtpa; Viraj; Mukand (long products only)	Melt chemistry, cleanliness, N control for AM350 (0.07–0.13%)
3	Hot roll	Reheat → roughing → Steckel or tandem finishing → anneal & pickle → HR coil	Same integrated mills	Jindal Stainless — Hisar Steckel 720k tpa, Hisar tandem 300k tpa, Jajpur 1.0 Mtpa (Siemens VAI)	Surface quality, HR gauge uniformity
4	**Cold roll to thin gauge**	20-Hi Sendzimir and 4-Hi mills with AGC; multi-pass reduction to 0.03–0.20 mm; bright/bell/pull-through annealing between passes	**Alleima** (to 0.015 mm at ±0.001 mm), Ulbrich, ATI, **TOKKIN**, Proterial, Aperam Imphy, voestalpine, Waelzholz, **Lamineries Matthey** (AM350 from 0.010 mm)	**Jindal Stainless SPD** 84,000 tpa to 0.076 mm; **IUP Jindal** 0.03–1.5 mm, 22,000 tpa; **Quality Foils** 0.10–4.0 mm; Hisar Metal; Singhal Strips	Thickness tolerance, flatness, grain structure for fatigue life
5	Temper & finish	Skin pass mill, tension leveller, strip grinding, bright annealing to 2R/BA, temper to spring hardness (¼H to full hard, HV530+)	Same precision mills; Iwatani sources spring and gasket tempers	Jindal SPD (strip grinding, skin pass, tension leveller); IUP Jindal (2 bright anneal lines, Brodeur leveller)	Hardness, flatness, surface cleanliness (2R/BA for vacuum)
6	**PRECISION SLITTING**	Rotary knife slitting of master coil to narrow strip; shimless tooling, computerised knife setting; edge conditioning — deburr, round, chamfer; burr height and direction specified	Alleima, Ulbrich, **Hempel Special Metals** (serves bellows makers, from 0.05 mm), Lamineries Matthey (±0.1 mm width on request). **★ Iwatani — Suzhou, Zhongshan, Thailand**	**IUP Jindal** — 3 Brodeur lines, shimless tooling, edge rounding, slit to 3.5 mm; **Quality Foils** — slit edge from 5.0 mm; **Jindal SPD** — precision slitters + 5 slitting lines in the CR complex; stockists Aesteiron, Sachiya	Width tolerance ±0.05 mm, burr ≤10% of thickness, camber, coil set
7A	**Formed / convoluted bellows**	Shear → roll to cylinder → **longitudinal seam weld** (TIG/plasma) → clean & inspect → telescope plies (2–20) → **hydroform** at 15–20k psi or mechanical/ mandrel form → neck weld, reinforcing rings	Witzenmann (HYDRA), Senior Flexonics, BOA Group, Belman, Flexider, EagleBurgmann, Hyspan, Penflex, Triad, Macoga, Comflex	Flexpert Bellows (2–20 ply), MB Metallic Bellows (100 NB to 8 m dia), Metallic Bellows (India), Witzenmann India, Fluidyne, Bhastrik, Triveni, Scutes India	Seam weld integrity, thinning control, EJMA / ASME VIII
7B	**Edge-welded bellows**	Blank/press diaphragm rings from foil → deburr → stack and nest → **ID weld → OD weld** (laser, micro-plasma, EB, TIG) → attach end pieces; nesting-ripple geometry for maximum stroke	**KSM Corp** (world's largest, ISO 6 cleanroom), Technetics BELFAB, MW Components/Servometer, Metal-Flex, Witzenmann, Irie Koken, MIRAPRO, Eagle Industry EKK, Valqua, BellowsTech	**★ Well Tech (Vasai)** 0.05–0.2 mm, AM350, He leak tested; **★ Fluidyne (Mysuru)** 0.05 mm diaphragms, racetrack UHV; Bhastrik 0.127–0.30 mm micro-plasma	Weld bead <0.1 mm, spring rate, cycle life 3M–10M cycles
8	Assembly & hardware	End plates, CF/KF/ISO flanges, liners, external covers, tie rods, gimbal	Bellows makers plus MW Components, Huntington,	**Fourvac** (XYZ manipulators, UHV valves, drives), **HHV** (Bengaluru), **APT** (UHV	Dimensional accuracy,

#	Stage	Process	Global players	India players	Technical gate
		and hinge hardware, braiding; sub-assembly into manipulators and translators	Nor-Cal, MDC Vacuum, Kurt J. Lesker	chambers, leak test to 1e-12 Torr·l/s), Metallic Bellows (India)	assembly cleanliness
9	**Clean, test & certify**	Ultrasonic and solvent clean → UHV bake → **ISO Class 5/6 cleanroom** assembly and packaging → **helium leak test** <1e-9 mbar·l/s → RGA certificate → fatigue/ cycle testing → double-bag	KSM (world's largest ISO 6 bellows cleanroom, ISO 5 final pack), Technetics (Class 100/1000), Witzenmann (UHV/UHP with optional RGA cert)	**⚠ No Indian facility found** with an ISO Class 5/6 bellows cleanroom or RGA capability. APT leak-tests to 1e-12 Torr·l/s; KASFAB has cleanrooms and precision welding	**★ This is the real barrier to semiconductor entry**
10	Component integration	Bellows built into the finished sellable component: vacuum valves, mechanical seals, feedthroughs, expansion joints, dampers, accumulators, MFCs, gas panels	**Valves:** VAT, SMC, Irie Koken, Ferrotec, Pfeiffer. **Seals:** EagleBurgmann, John Crane, Flowserve, Eagle Industry, Chesterton. **Chambers:** MIRAPRO	Well Tech supplies Indian mechanical seal makers; Fourvac (UHV valves); HHV, APT (chambers); EPC and piping contractors for expansion joints	Valve cycle life 3M+, leak rate, particle generation
11	OEM / equipment build	Component goes into capital equipment: wafer fab tools, aero systems, pumps and compressors, vehicles, electrolysers, process plant	**Semi WFE:** Applied Materials, Lam Research, Tokyo Electron, ASML, KLA, Screen, Kokusai, YES. **Aero:** Boeing, Airbus, GE, Safran. **Pumps:** KSB, Sulzer, Grundfos	**★ KASFAB Tools (Doddaballapur)** — India's first semiconductor equipment contract manufacturer, for AMAT, Lam, TEL, YES; AMAT India (Bengaluru); HAL, BHEL, Kirloskar	Tool qualification, OEM approved-vendor-list entry (12–24 months)
12	End user & aftermarket	Installed and operated; bellows then **re-ordered as a consumable** — 3M cycle service life, user-replaceable; recurring MRO revenue for decades	**Fabs:** TSMC, Samsung, Intel, Micron, SK Hynix. **Refineries:** Shell, Exxon. **Space:** NASA, ESA, JAXA. **Fusion:** ITER, accelerators	**Fabs/OSAT:** Micron Sanand, Tata-PSMC Dholera, TSAT Assam, Kaynes, CG Semi. **Refinery:** IOCL, RIL, BPCL. **Power:** NTPC, NPCIL. **Space/defence:** ISRO, BARC, DRDO. **Steel:** SAIL, Tata, JSW	**★ Aftermarket is recurring, higher margin, and fab-located**

6. Five things the map makes visible

1. Iwatani and Jindal together already span stages 1 to 6 — but not in the same grades or the same places. Jindal covers 1–5 in India in austenitic, ferritic, martensitic and duplex. Iwatani covers 5–6 in Japan, China and Thailand, in spring, gasket and ultrathin grades down to 8 µm including 631. The combination is genuinely complementary. What neither covers is **AM350 melted anywhere**, which sits at stage 2.

2. The proposed entry point at stage 6 is already occupied in India. Both IUP Jindal and Quality Foils precision-slit and explicitly name bellows as a served application. So "we can slit precisely" is not a differentiated pitch. The gap sits **upstream at stages 2 and 4** — grade availability and tolerance class — which is a more defensible position anyway, because it is harder to replicate.

3. Stage 9 is the real barrier to semiconductor entry, and it is not a materials problem. Cleanroom assembly, helium leak certification and RGA capability are what separate an Indian bellows maker from a semiconductor-qualified one. No Indian facility was found with it. **Iwatani cannot supply this**, which is precisely why the strategy stops at material supply rather than promising a semiconductor components business.

4. Margin rises monotonically downstream — and so does the entry barrier. Commodity melting and hot rolling at 10–11% indicative gross margin; precision slitting ~25%; edge-welded manufacture ~42%; cleanroom qualification and component integration ~48–52%. Moving from stage 2–3 economics to stage 5–6 economics roughly **doubles margin without requiring a manufacturing licence, a cleanroom, or OEM approval.** That is the realistic ambition. Stages 9–12 are where the money is and where a trading house cannot credibly go.

5. The MRO loop is the most under-appreciated feature of the chain. Bellows are a designed-in consumable — SMC and VAT both rate their valves at 3 million cycles with user-replaceable bellows, and Irie Koken quotes 10,000 to 10 million cycles. So demand re-enters at stage 10 for the life of every tool ever installed. That replacement stream is recurring, higher-margin, and located **at the fab** rather than at the tool builder — which is the one part of the semiconductor opportunity that India's fab build-out actually does create.

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7. How to use these charts

Audience	Use
Internal Iwatani approval	Chart 29 alone. It shows in one image why the Stainless Steel Division should move from stage 2–3 economics to stage 5–6, and it does so without over-promising a components business
Jindal Stainless discussion	Charts 27 and 29. Positions Jindal at stages 1–5, identifies the missing grade at stage 2, and shows what the joint position could be worth
Indian bellows maker (Well Tech, Fluidyne)	Chart 28. Puts them at stage 7B, shows what they must import, and frames Iwatani as the fix for stage 4–6 rather than a competitor
Monthly internal meeting	All three, as the standing industry-structure reference. Update the India player boxes as the customer discovery in `08` confirms or removes names

Caveat to state whenever chart 29 is shown: the margin percentages are our indicative estimates, not measured data. The shape of the curve is well supported by the structure of the industry; the specific numbers are a prioritisation aid. Anyone asking for the source of "42%" should be told it is a judgement, not a citation.

Chart index

Chart	File	Purpose
27	`27-valuechain-upstream.png`	Stages 1–6, ore to precision slit strip, with players and gates
28	`28-valuechain-downstream.png`	Stages 7–12, bellows manufacture to end user and MRO loop
29	`29-value-pool.png`	Indicative margin by stage, with Iwatani and Jindal positions marked

Generated by `charts/make_valuechain_charts.py`. Player lists are drawn from the sources in `06-source-register.md`, `08-supply-chain-sourcing.md` and `07-research-reconciliation.md`.